Semantic File Searching

SPL 1 Project Proposal

Presented by

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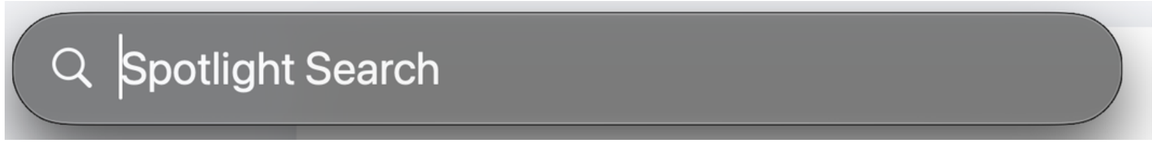
Lecturer

Institute of Information Technology (IIT)

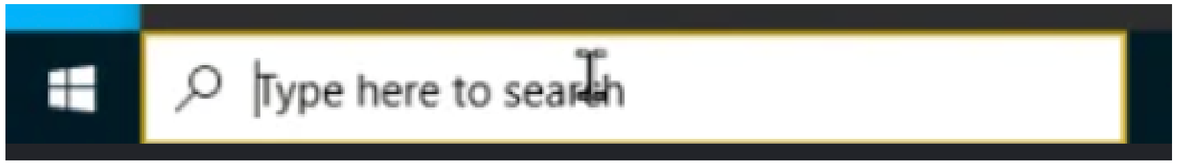
University of Dhaka

We have all used this

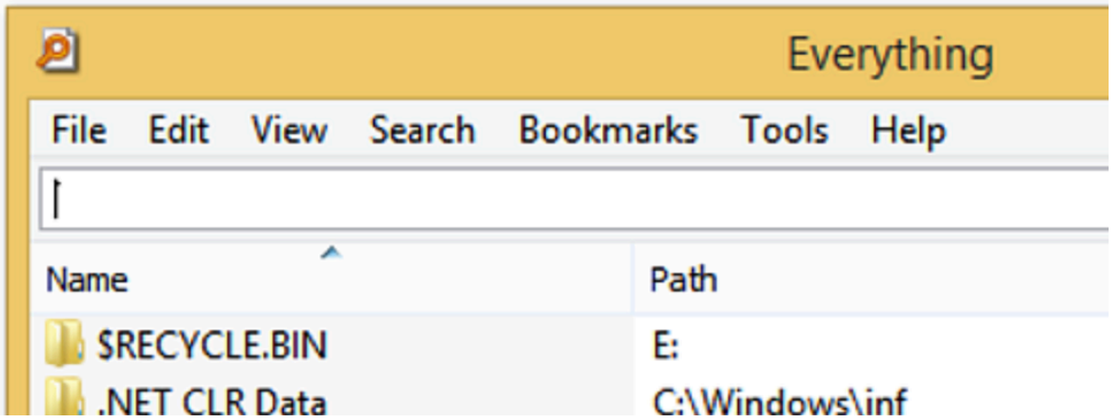
Macos



Windows



Third party



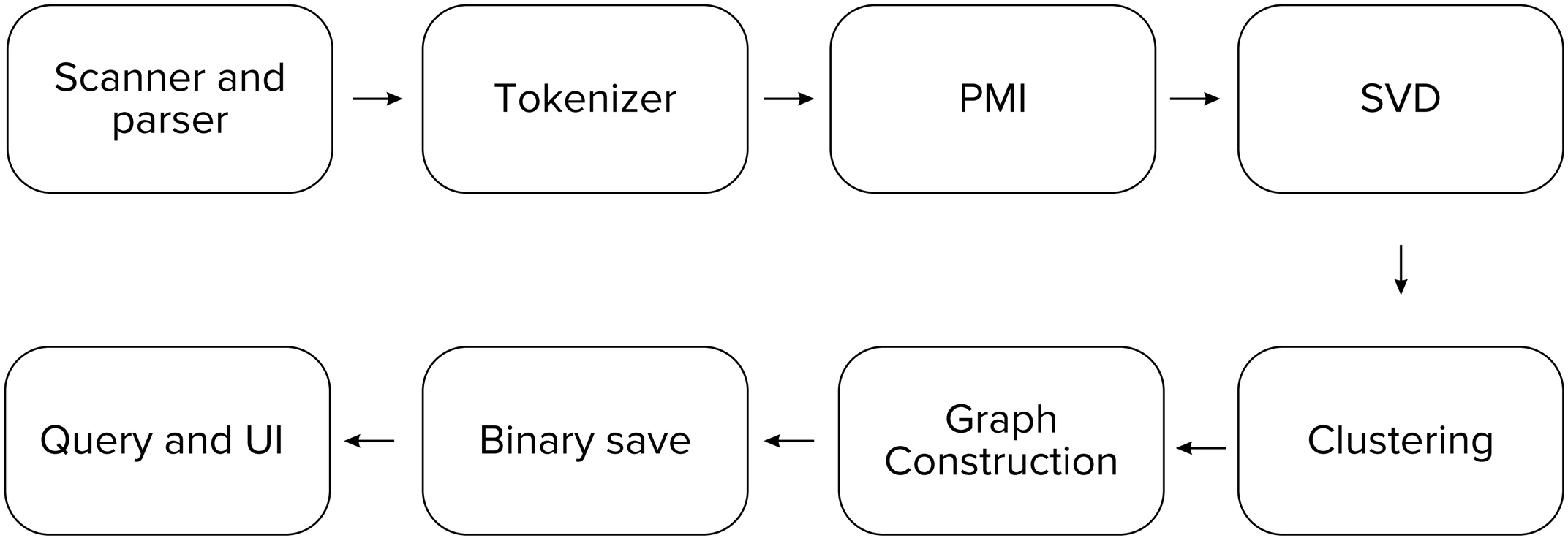
What if

> where is my note on teaching computers to understand human speech

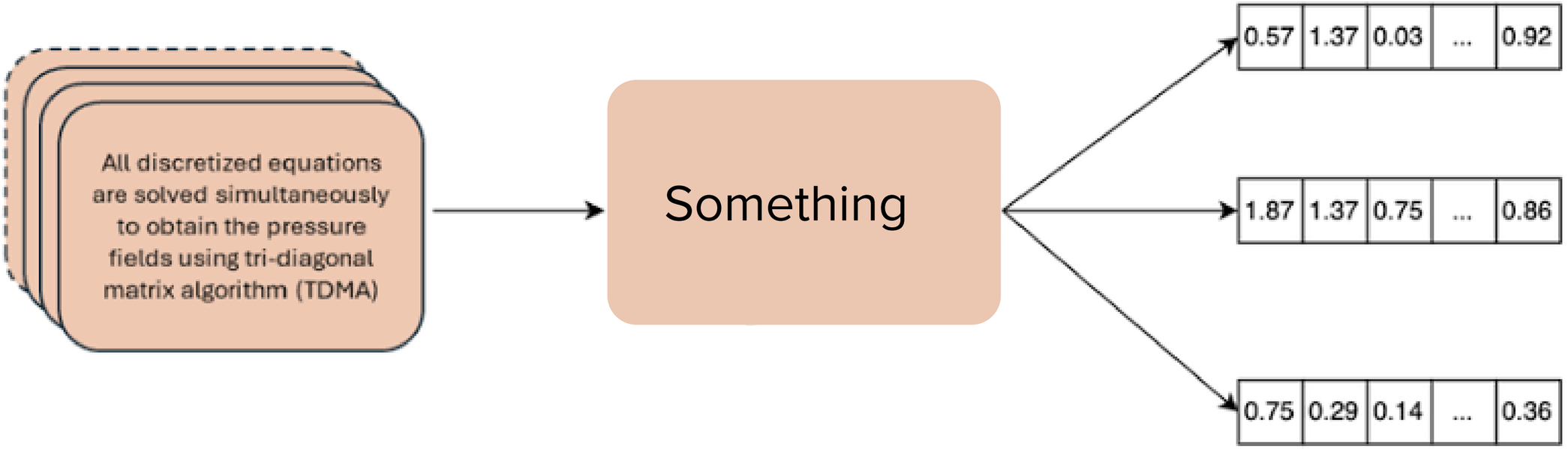
→ nlp.txt

→ note22september.pdf

Components


Scanner and parser
Tokenizer
PMI
SVD
Clustering
Graph Construction
Binary save
Query and UI

Language and Embeddings

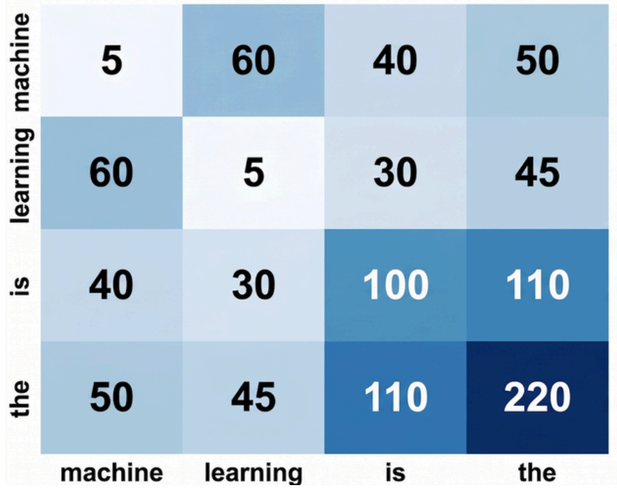


Tokenization (Byte Pair Encoding)

C++[c] is a high-level, general-purpose programming language created by Danish computer scientist Bjarne Stroustrup.

34, 4880, 58, 66, 60, 318, 257, 1029, 12, 5715, 11, 2276, 12, 29983, 8300, 3303, 2727, 416, 20849, 3644, 11444, 347, 9491, 710, 520, 472, 301, 12618, 13, 220

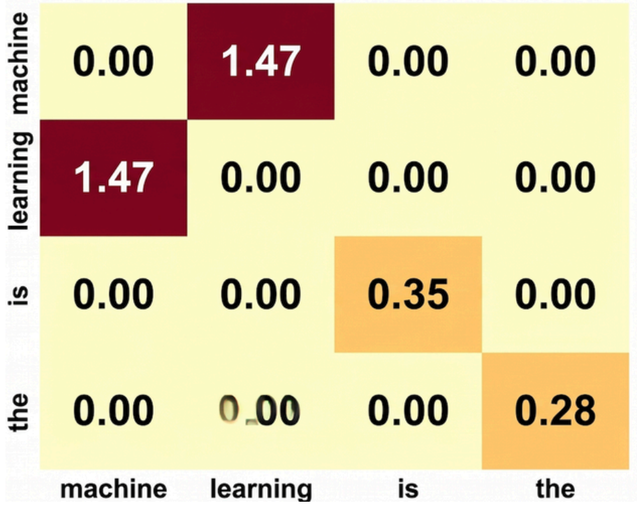
Co occurence Matrix



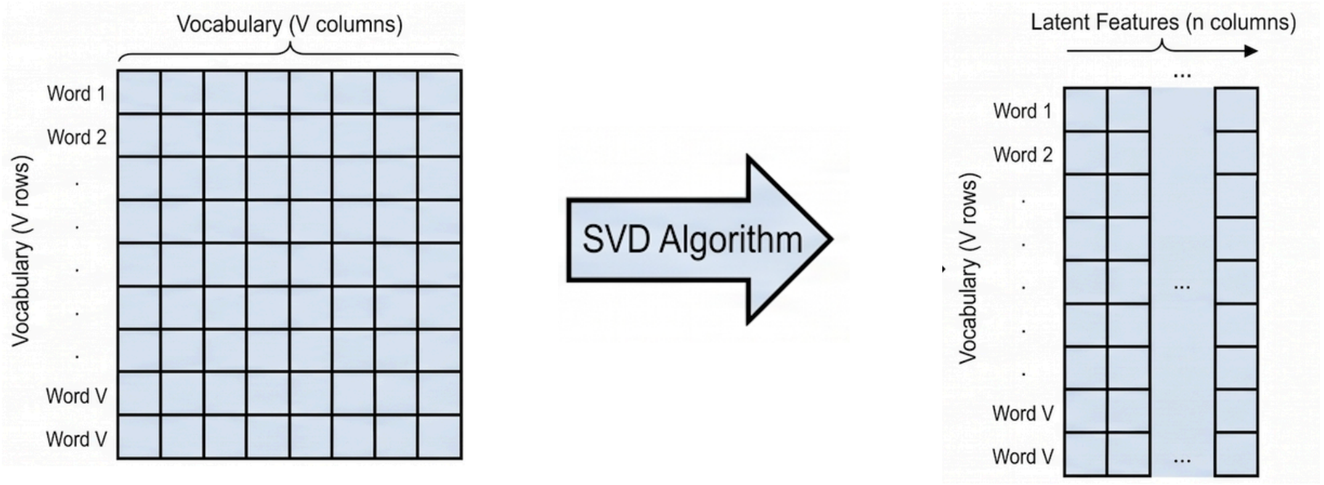
Pointwise Mutual Information

$$\mathrm{PMI}(w,c) = \log_2 \frac{P(w,c)}{P(w)P(c)}$$

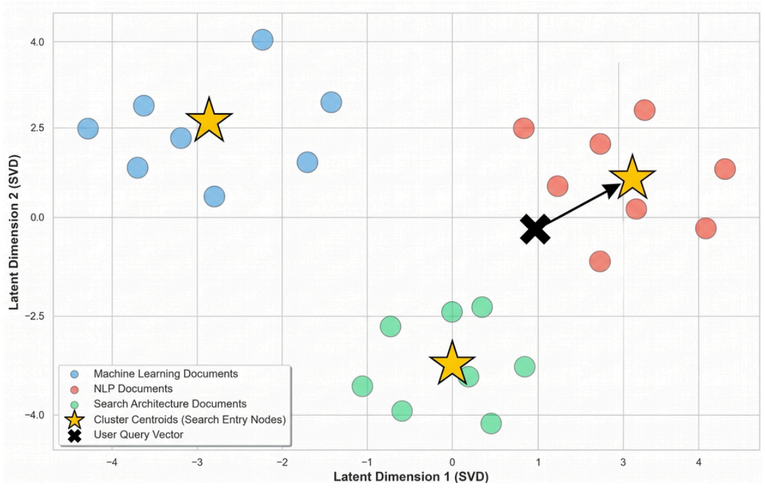
$$\mathrm{PPMI}(w,c) = \max(\log_2 \frac{P(w,c)}{P(w)P(c)}, 0)$$

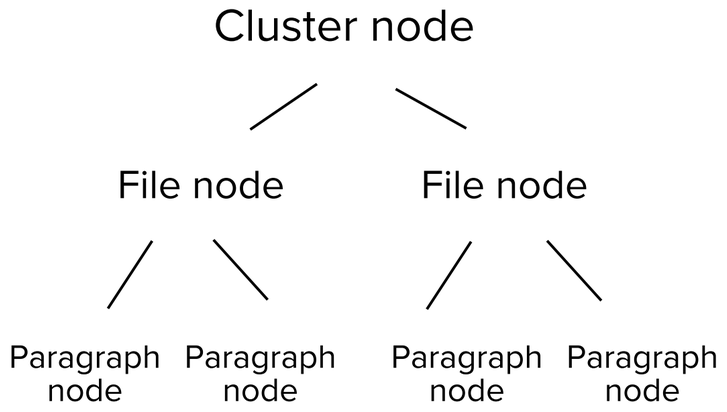


Singular Value Decomposition (SVD)

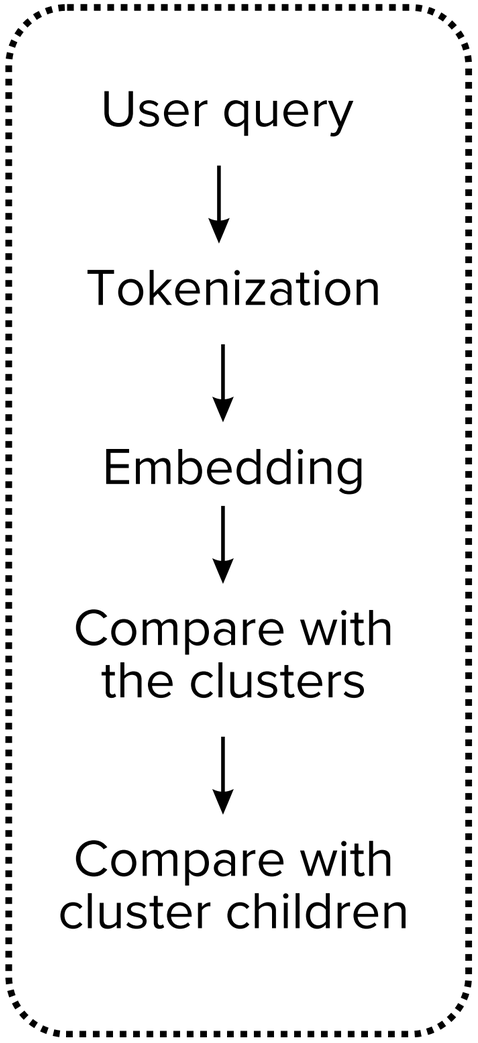


K means clustering and graph

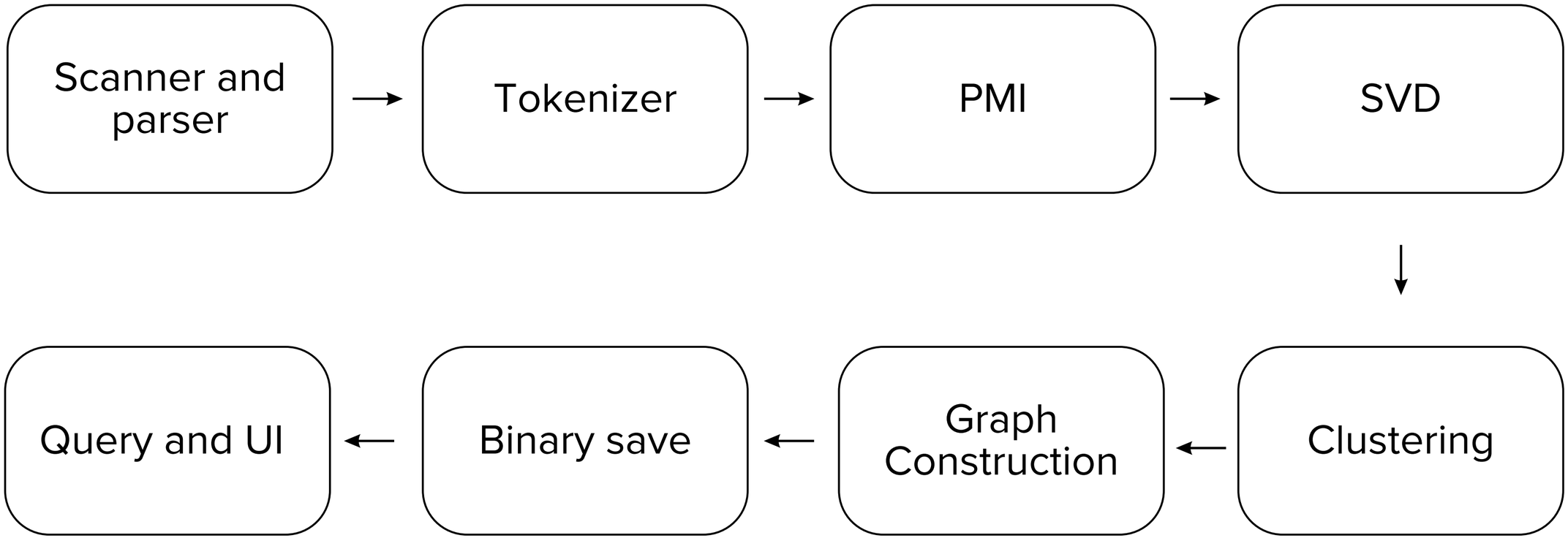




Query to Result



Components Revisited



Target Users

Any personal computer user

Technologies

- C++
- External libraries for pdf to text conversion and graph visualization
- Git, Github

QnA

Thank you